WORKED SOLUTION

STEP III 1994, Question 9

Mechanics • Tautochrone and Collisions

STEP III 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q9

This question has a full guided treatment in *Beyond STEP*, Section 11.4, *The Tautochrone*: https://beyondstep.keithmcnulty.org/ch11/tautochrone.html

The key idea

Energy conservation along the surface gives a first integral, and the shape of the bowl, $y = s^2/(4k)$, turns it into the energy equation of simple harmonic motion in the arc-length s. The period of simple harmonic motion doesn't depend on the amplitude, so every particle released from rest reaches V after the same time, wherever it starts. Because of this, the two particles in the final part are certain to meet at V, and they meet again and again.

Solution

The constant

We model the particle as a point mass on a smooth surface. Two forces act on it: its weight and the normal reaction. The particle starts from rest, so its velocity starts with no horizontal component perpendicular to the vertical plane through its starting point and the axis. Neither force has a component perpendicular to that plane (the reaction of an axially symmetric surface lies in the plane through the axis), so the particle stays in that vertical cross-section. It moves along the curve $s = 2\sqrt{ky}$, with speed $|\mathrm{d}s/\mathrm{d}t|$.

The surface is smooth, so the normal reaction is perpendicular to the velocity and does no work. Only gravity does work, so mechanical energy is conserved:

$$\tfrac{1}{2}m\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + mgy = \text{constant}.$$

Multiplying by $2/m$ shows that $\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + 2gy$ is constant, as required. Released from rest at height h, the constant is $2gh$.

The time to reach V

On the curve, $y = s^2/(4k)$. Substituting,

$$\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + \frac{g}{2k}s^2 = 2gh.$$

Differentiating with respect to t and dividing by $2\,\mathrm{d}s/\mathrm{d}t$ (valid while the particle is moving) gives

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{g}{2k}s,$$

which is simple harmonic motion in s with $\omega^2 = \dfrac{g}{2k}$, centred on $s = 0$, the point V.

Ask yourself Why is the time from release to V a quarter of a period?

The particle starts from rest at the extreme position $s_0 = 2\sqrt{kh}$, so $s = s_0 \cos \omega t$. It first reaches $s = 0$ when $\omega t = \frac{\pi}{2}$, a quarter of the period $2\pi/\omega$. So the time to reach V is

$$T = \frac{\pi}{2\omega} = \frac{\pi}{2}\sqrt{\frac{2k}{g}} = \pi\sqrt{\frac{k}{2g}},$$

as required. This doesn't depend on h: the bowl is a *tautochrone*.

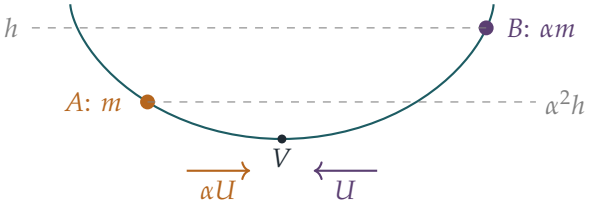
For later use, the speed at V is ωs_0, or directly from the energy equation with $y = 0$, $\sqrt{2gh}$.

The two particles

Treat the two particles as point masses that collide directly at V. The tangent there is horizontal, so the impact is along a horizontal line. Call the particle of mass m particle A (released at height $\alpha^2 h$) and the particle of mass αm particle B (released at height h on the opposite side).

They meet at V. By the previous part, each particle reaches V after time T, whatever its starting height. They are released together, so they arrive at V together and collide there at time T.

Speeds just before impact. Write $U = \sqrt{2gh}$. From the energy equation, A arrives with speed $\sqrt{2g\alpha^2 h} = \alpha U$ and B with speed U, travelling in opposite directions. Take the direction of A's motion as positive.



Momentum before impact is

$$m(\alpha U) + \alpha m(-U) = 0.$$

This is why the masses and heights were chosen as they were.

After the first impact. Let the velocities afterwards be v_A and v_B. Conservation of momentum and Newton's experimental law give

$$mv_A + \alpha m\, v_B = 0, \qquad v_B - v_A = \alpha\big(\alpha U - (-U)\big) = \alpha(1+\alpha)U.$$

From the first equation $v_B = -v_A/\alpha$. Substituting into the second gives $-v_A(1+\alpha)/\alpha = \alpha(1+\alpha)U$, so

$$v_A = -\alpha^2 U, \qquad v_B = \alpha U.$$

Both velocities are reversed and multiplied by α. (With zero total momentum, the relative velocity and each individual velocity are all scaled by $-\alpha$.) So each particle goes back up its *own* side of the bowl. By energy conservation, A rises to height $\alpha^4 h$ and B to height $\alpha^2 h$.

The pattern repeats. Each particle comes to rest at its new highest point at the same moment, a time T after the impact. Each then takes a further time T to return to V. So they collide again at V at time $3T$, with A from height $\alpha^4 h$ and B from height $\alpha^2 h$. This is exactly the starting configuration with h replaced by $\alpha^2 h$. Again the total momentum is zero, and again both velocities are reversed and multiplied by α.

By induction, the nth collision happens at V at time $(2n-1)T$, and after it

$$A \text{ rises to height } \alpha^{2n+2}h, \qquad B \text{ rises to height } \alpha^{2n}h,$$

each on the side of the bowl it started from. The particles oscillate on their own sides, with collisions at V at the equally spaced times $T, 3T, 5T, \ldots$. At each collision the speeds drop by the factor α, so the heights drop by the factor α^2 and a fraction $1-\alpha^2$ of the kinetic energy is lost. The collisions never stop, because the gaps between them are always $2T$. The amplitudes decrease geometrically, and both particles tend to rest at V as $t \to \infty$ without ever reaching it in finite time.

Checks

- Dimensions: k is a length (since $s^2 = 4ky$), so $\sqrt{k/g}$ is a time.
- Numerically, $\int_0^{s_0} \mathrm{d}s/\sqrt{2g(h-y)}$ was evaluated by quadrature for random k and h. With $k=1$, $g=9.81$ it gives $0.7092\ldots = \pi\sqrt{1/19.62}$, independent of h.
- We stepped the motion of both particles numerically (with $\alpha = 0.6$ and $\alpha = 0.3$) and applied the impact law at each meeting. The collisions came at $t/T = 1, 3, 5, 7$, always at V and with zero total momentum. After the first collision with $\alpha = 0.6$, the heights were $0.1296h = \alpha^4 h$ and $0.36h = \alpha^2 h$.
- Limiting case $\alpha \to 1$: equal masses from equal heights, perfectly elastic, so each bounces back to its starting height for ever, as expected.

Common slips

- Taking V as half a period from the start: the particle starts at an extreme of the oscillation, so V is a quarter-period away.
- Forgetting the reversal of direction in Newton's law: take the velocities with signs, not just speeds.
- Claiming the particles come to rest after finitely many collisions. The interval between collisions is constant, not shrinking (unlike a ball bouncing on a floor), so there are infinitely many collisions spread over infinite time.

Beyond STEP

The intrinsic equation $s^2 = 4ky$ describes a cycloid. This question rediscovers Huygens' tautochrone: on an inverted cycloid, the time to slide to the bottom doesn't depend on the starting point. Huygens used this to design pendulum clocks whose period doesn't depend on the swing. The same curve is the brachistochrone, the curve of fastest descent, found with the calculus of variations, a standard topic in second-year undergraduate mechanics. For more on oscillations and collisions, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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